

Лабораторная работа №3

Дисциплина: Сетевые технологии

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0.1 Цели и задачи

- Изучение посредством Wireshark кадров Ethernet, анализ PDU протоколов транспортного и прикладного уровней стека TCP/IP.

1. Выполнение лабораторной работы

2. MAC-адресация

2.1 ipconfig

```
C:\Users\bahis>ipconfig

Windows IP Configuration

Ethernet adapter Ethernet:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Ethernet adapter Ethernet 9:

    Connection-specific DNS Suffix  . :
    Link-local IPv6 Address . . . . . : fe80::1cf3:488b:3e25:51ce%31
    IPv4 Address. . . . . : 192.168.56.1
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . :

Wireless LAN adapter Local Area Connection* 1:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 13:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Wi-Fi:

    Connection-specific DNS Suffix  . :
    Link-local IPv6 Address . . . . . : fe80::1d20:6002:df86:b816%22
    IPv4 Address. . . . . : 172.16.58.120
    Subnet Mask . . . . . : 255.255.254.0
    Default Gateway . . . . . : 172.16.58.1

Ethernet adapter vEthernet (Default Switch):

    Connection-specific DNS Suffix  . :
    Link-local IPv6 Address . . . . . : fe80::b15c:fc1b:fc1b:3462%62
    IPv4 Address. . . . . : 172.25.224.1
    Subnet Mask . . . . . : 255.255.240.0
    Default Gateway . . . . . :
```

Рисунок 1: Команда ipconfig

2.2 ipconfig /all

```
Wireless LAN adapter Wi-Fi:  
  
    Connection-specific DNS Suffix  . :  
    Link-local IPv6 Address . . . . . : fe80::1d20:6002:df86:b816%22  
    IPv4 Address. . . . . : 172.16.58.120  
    Subnet Mask . . . . . : 255.255.254.0  
    Default Gateway . . . . . : 172.16.58.1
```

Рисунок 2: Команда ipconfig /all

3. Анализ кадров канального уровня в Wireshark

3.1 Wireshark

```
Connection-specific DNS Suffix . :  
Description . . . . . : Intel(R) Dual Band Wireless-N 7260  
Physical Address. . . . . : F8-16-54-69-79-63  
DHCP Enabled. . . . . : Yes  
Autoconfiguration Enabled . . . . : Yes  
Link-local IPv6 Address . . . . : fe80::1d20:6002:df86:b816%22(Preferred)  
IPv4 Address. . . . . : 172.16.58.120(Preferred)  
Subnet Mask . . . . . : 255.255.254.0  
Lease Obtained. . . . . : Monday, December 15, 2025 6:19:29 PM  
Lease Expires . . . . . : Monday, December 15, 2025 8:19:29 PM  
Default Gateway . . . . . : 172.16.58.1  
DHCP Server . . . . . : 192.168.80.59  
DHCPv6 IAID . . . . . : 150476372  
DHCPv6 Client DUID. . . . . : 00-01-00-01-2B-15-97-11-34-64-A9-7A-40-9B  
DNS Servers . . . . . : 192.168.150.37  
                        192.168.150.36  
NetBIOS over Tcpi. . . . . : Enabled
```

Рисунок 3: Запуск захвата трафика

3.2 Пинг

1	0.000000	172.16.58.120	64.233.163.188	TCP	55 52460 → 5228 [ACK] Seq=
2	0.020318	64.233.163.188	172.16.58.120	TCP	66 5228 → 52460 [ACK] Seq=
3	0.225313	172.16.59.51	224.0.0.251	MDNS	459 Standard query 0x0000 F
4	0.345090	172.16.58.120	149.154.167.99	TLSv1.2	253 Application Data
5	0.389977	149.154.167.99	172.16.58.120	TCP	60 443 → 54720 [ACK] Seq=1
6	0.392308	149.154.167.99	172.16.58.120	TLSv1.2	167 Application Data
7	0.436070	172.16.58.120	149.154.167.99	TCP	54 54720 → 443 [ACK] Seq=2
8	0.531951	172.16.58.45	224.0.0.251	MDNS	108 Standard query 0x0000 T
9	0.739166	172.16.59.57	224.0.0.251	MDNS	486 Standard query response
10	0.739166	Apple_c2:87:f9	Broadcast	ARP	60 Who has 172.16.58.107?

Рисунок 4: Пинг шлюза по умолчанию

3.3 Фильтр ICMP

```
C:\Users\bahis>
C:\Users\bahis>ping 172.16.58.1

Pinging 172.16.58.1 with 32 bytes of data:
Reply from 172.16.58.1: bytes=32 time=1ms TTL=254
Reply from 172.16.58.1: bytes=32 time=9ms TTL=254
Reply from 172.16.58.1: bytes=32 time=10ms TTL=254
Reply from 172.16.58.1: bytes=32 time=2ms TTL=254

Ping statistics for 172.16.58.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 10ms, Average = 5ms
```

Рисунок 5: Кадр ICMP - эхо-запрос: информация о длине кадра, типе Ethernet и MAC-адресах

3.4 Фильтр ICMP

35	0.409663	CenturyXinya_d9:95:...	Broadcast	ARP	60	Who has 169.254.169.254?
80	1.123797	c6:61:f0:f9:d0:f0	Broadcast	ARP	42	Who has 172.16.58.1? Tel
82	1.328783	CenturyXinya_d9:95:...	Broadcast	ARP	60	Who has 169.254.169.254?
111	2.251081	CenturyXinya_d9:95:...	Broadcast	ARP	60	Who has 169.254.169.254?
146	4.300060	12:8f:f0:1a:15:1d	Broadcast	ARP	60	Who has 172.16.58.1? Tel
148	4.300060	Intel_be:c5:6c	Broadcast	ARP	60	Who has 172.16.58.21? Te
239	6.676289	Intel_be:c5:6c	Broadcast	ARP	60	Who has 172.16.58.51? Te

Рисунок 6: Кадр ICMP - эхо-ответ: информация о длине кадра, типе Ethernet, MAC-адресах

3.5 Фильтр ARP

```
[Time delta from previous displayed frame: 2.376229000 s  
[Time since reference or first frame: 6.676289000 second  
Frame Number: 239  
Frame Length: 60 bytes (480 bits)  
Capture Length: 60 bytes (480 bits)  
[Frame is marked: False]  
[Frame is ignored: False]  
[Protocols in frame: eth:ethertype:arp]  
Character encoding: ASCII (0)  
[Coloring Rule Name: ARP]  
[Coloring Rule String: arp]
```

Рисунок 7: Кадр ARP: информация о длине кадра, типе Ethernet, MAC-адресах

3.6 Пинг

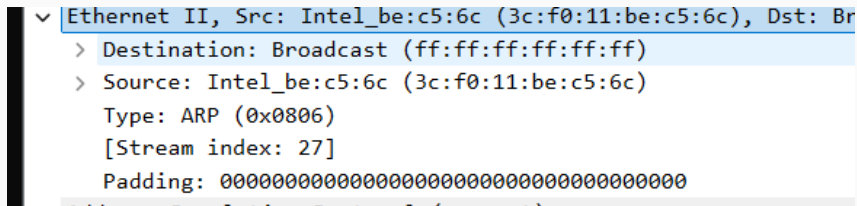


Рисунок 8: Пинг сайта www.yandex.ru

3.7 Фильтр ICMP

```
C:\Users\bahis>ping www.yandex.ru

Pinging www.yandex.ru [77.88.44.55] with 32 bytes of data:
Reply from 77.88.44.55: bytes=32 time=5ms TTL=56
Reply from 77.88.44.55: bytes=32 time=4ms TTL=56
Reply from 77.88.44.55: bytes=32 time=3ms TTL=56
Reply from 77.88.44.55: bytes=32 time=4ms TTL=56

Ping statistics for 77.88.44.55:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 3ms, Maximum = 5ms, Average = 4ms
```

Рисунок 9: Запрос протокола ICMP

3.8 Фильтр ICMP

No.	Time	Source	Destination	Protocol	Length	Info
→ 6379	211.452484	172.16.58.120	77.88.44.55	ICMP	74	Echo (ping) request id=0x0001, seq=5/1280, ttl=128 (
← 6380	211.457316	77.88.44.55	172.16.58.120	ICMP	74	Echo (ping) reply id=0x0001, seq=5/1280, ttl=56 (r
6409	212.465927	172.16.58.120	77.88.44.55	ICMP	74	Echo (ping) request id=0x0001, seq=6/1536, ttl=128 (
6410	212.470021	77.88.44.55	172.16.58.120	ICMP	74	Echo (ping) reply id=0x0001, seq=6/1536, ttl=56 (r
6455	213.484607	172.16.58.120	77.88.44.55	ICMP	74	Echo (ping) request id=0x0001, seq=7/1792, ttl=128 (
6456	213.487931	77.88.44.55	172.16.58.120	ICMP	74	Echo (ping) reply id=0x0001, seq=7/1792, ttl=56 (r
6513	214.501521	172.16.58.120	77.88.44.55	ICMP	74	Echo (ping) request id=0x0001, seq=8/2048, ttl=128 (
6514	214.505791	77.88.44.55	172.16.58.120	ICMP	74	Echo (ping) reply id=0x0001, seq=8/2048, ttl=56 (r

> Frame 6379: Packet, 74 bytes on wire (592 bits), 74 bytes captured (584 bits) on interface 0	0000	70 18 a7 60 9c c5 f8 16 54 69 79 63 08 00 45 00	p...T
▼ Ethernet II, Src: Intel_69:79:63 (f8:16:54:69:79:63), Dst: Intel_69:79:63 (f8:16:54:69:79:63)	0010	00 3c 0b 6b 00 00 80 01 cf 3e ac 10 3a 78 4d 58	.<.k...
Destination: Cisco_60:9c:c5 (70:18:a7:60:9c:c5)	0020	2c 37 08 00 4d 56 00 01 00 05 61 62 63 64 65 66	,7..MV...
...0... = LG bit: Globally unique address	0030	67 68 69 6a 6b 6c 6d 6e 6f 70 71 72 73 74 75 76	ghijklmn o
...0... = IG bit: Individual address	0040	77 61 62 63 64 65 66 67 68 69	wabcdefgh
Source: Intel_69:79:63 (f8:16:54:69:79:63)			
...0... = LG bit: Globally unique address			
...0... = IG bit: Individual address			
Type: IPv4 (0x0800)			
[Stream index: 13]			
> Internet Protocol Version 4, Src: 172.16.58.120, Dst: 77.88.44.55			

Рисунок 10: Ответ протокола ICMP

3.9 Фильтр ICMP

icmp						
No.	Time	Source	Destination	Protocol	Length	Ir
→ 6379	211.452484	172.16.58.120	77.88.44.55	ICMP	74 E	
← 6380	211.457316	77.88.44.55	172.16.58.120	ICMP	74 E	
6409	212.465927	172.16.58.120	77.88.44.55	ICMP	74 E	
6410	212.470021	77.88.44.55	172.16.58.120	ICMP	74 E	
6455	213.484607	172.16.58.120	77.88.44.55	ICMP	74 E	
6456	213.487931	77.88.44.55	172.16.58.120	ICMP	74 E	
6513	214.501521	172.16.58.120	77.88.44.55	ICMP	74 E	
6514	214.505791	77.88.44.55	172.16.58.120	ICMP	74 E	

> Frame 6379: Packet, 74 bytes on wire (592 bits), 74 bytes captured (592 bits)

▼ Ethernet II, Src: Intel_69:79:63 (f8:16:54:69:79:63), Dst: Cisco_60:9c:c5 (78:0c:cc:60:9c:c5)

> Destination: Cisco_60:9c:c5 (78:0c:cc:60:9c:c5)

> Source: Intel_69:79:63 (f8:16:54:69:79:63)

Type: IPv4 (0x0800)

[Stream index: 13]

> Internet Protocol Version 4, Src: 172.16.58.120, Dst: 77.88.44.55

▼ Internet Control Message Protocol

Type: Echo (ping) request (8)

Code: 0

Checksum: 0x4d56 [correct]

[Checksum Status: Good]

Рисунок 11: Кадр ARP - эхо-ответ

4. Анализ протоколов транспортного уровня в Wireshark

4.1 http://info.cern.ch/

The screenshot displays a Wireshark interface with a network capture of an HTTP GET request. The top pane shows a list of packets, with packet 16760 selected. The middle pane shows the details of the selected packet, and the bottom pane shows the raw hex data.

Packet List:

No.	Time	Source	Destination	Protocol	Length	Info
16760	560.625660	172.16.58.120	151.101.206.172	HTTP	356	GET /msdownload/update/v3/static/trusted/en/autorootstl.cab7b6fac61f4332af3 HTTP/1.1
16762	560.669377	151.101.206.172	172.16.58.120	HTTP	256	HTTP/1.1 304 Not Modified
18201	604.896218	172.16.58.120	146.75.122.172	HTTP	356	GET /msdownload/update/v3/static/trusted/en/pinrulesstl.cab721dc3ced9a58cd5 HTTP/1.1
18211	605.141490	146.75.122.172	172.16.58.120	HTTP	256	HTTP/1.1 304 Not Modified
18211	605.154737	172.16.58.120	146.75.122.172	HTTP	342	GET /msdownload/update/v3/static/trusted/en/diallowdcertstl.cab751d37c42f886bcd3 HTTP/1.1
18217	605.203744	146.75.122.172	172.16.58.120	HTTP	256	HTTP/1.1 304 Not Modified

Packet Details:

- Urgent Pointer: 0
- [Timestamps]
- [SEQ/ACK analysis]
- [Client Contiguous Streams: 1]
- [Server Contiguous Streams: 1]
- TCP payload (282 bytes)
- Hypertext Transfer Protocol
 - GET /msdownload/update/v3/static/trusted/en/autorootstl.cab7b6fac61f4332af3 HTTP/1.1\r\n
 - Request Method: GET
 - Request URI: /msdownload/update/v3/static/trusted/en/autorootstl.cab7b6fac61f4332af3
 - Request URI Path: /msdownload/update/v3/static/trusted/en/autorootstl.cab
 - Request URI Query: b6fac61f4332af3
 - Request Version: HTTP/1.1
 - Connection: Keep-Alive\r\n
 - Accept: */*\r\n
 - If-Modified-Since: Thu, 28 Aug 2025 21:00:38 GMT\r\n
 - If-None-Match: *48ea0c5e18d1c10\r\n
 - User-Agent: Microsoft-CryptoAPI/10.0\r\n

Raw Data (Hex):

```
0000 70 18 a7 60 9c c5 f8 16 5a 69 79 63 08 00 45 00 p.....Tlyc-E-
0010 01 42 8f 7e 40 00 00 06 1d 9d ac 10 3a 78 97 65 -B-m.....ix.e
0020 4e ac 4e 03 00 50 ed 37 8f 69 42 52 2f 59 50 18 ...P-7-LBVP.
0030 62 03 78 00 00 00 47 45 54 20 2f 66 73 64 6f 79 ..x-GE T/msdo
0040 0e 0c 6f 61 64 2f 75 70 64 61 74 65 1f 76 33 2f ..load/up date/v3/
0050 73 74 61 74 60 61 2f 74 72 75 73 74 65 64 72 2f ..static/ trusted/
0060 05 0a 2f 61 35 3a 60 72 6f 6f 74 74 6c 2a 63 ..p/autorootstl.c
0070 61 6f 3f 62 36 66 36 61 63 36 31 66 34 33 33 32 ..b6fac61f4332
0080 61 66 33 20 40 54 54 50 2f 31 2a 31 6d 0a 43 6f ..af3 HTTP /1.1-Co
0090 6e 6e 65 63 74 69 6f 6e 3a 20 4b 65 65 70 2d 43 ..nection : Keep-A
00a0 6c 69 76 65 0d 0a 41 63 63 65 70 74 3a 20 2a 2f ..lve-Ac cept: */
00b0 2a 08 0a 49 66 2d 4d 6f 64 09 66 09 65 64 2d 53 ..*.If-Mod ified-S
00c0 69 6e 63 65 3a 20 54 68 75 2c 20 32 38 20 41 75 ..lce: Th u, 28 Au
00d0 67 20 32 30 32 35 20 32 31 3a 30 30 3a 33 38 20 ..g 2025 2 1:00:38
00e0 47 4d 54 0a 0a 49 66 2d 4e 6f 6e 65 2d 4d 61 74 ..GMT-If: None-Mat
00f0 63 68 3a 20 22 34 38 65 61 30 63 65 35 65 31 ..ch: "48 ea0c5e1
0100 38 64 63 31 3a 30 22 0a 0a 55 73 65 72 2d 41 67 ..8d c10" -User-Ag
0110 65 0a 74 3a 20 4d 69 63 72 6f 73 6f 66 74 2d 43 ..ent: Mic rsoft-C
0120 72 79 70 74 6f 41 50 49 2f 31 30 2a 30 0d 0a 68 ..ryptoAPI /10.0-
0130 6f 73 74 3a 30 63 74 6c 64 6c 2a 77 69 6e 64 6f ..ort: (S) dLwindo
0140 77 73 75 70 64 61 74 65 2a 63 6f 6d 0d 0a 0d ..wupdate .com:....
```

Рисунок 12: Кадр http - запрос

4.2 http://info.cern.ch/

http						
No.	Time	Source	Destination	Protocol	Length	Info
→ 16740	560.625660	172.16.58.120	151.101.206.172	HTTP	336	GET /m
← 16742	560.669377	151.101.206.172	172.16.58.120	HTTP	256	HTTP/1
18201	604.806218	172.16.58.120	146.75.122.172	HTTP	336	GET /m
18211	605.141490	146.75.122.172	172.16.58.120	HTTP	256	HTTP/1
18213	605.154737	172.16.58.120	146.75.122.172	HTTP	342	GET /m
18217	605.203744	146.75.122.172	172.16.58.120	HTTP	256	HTTP/1

Source Port: 52739
Destination Port: 80
[Stream index: 53]
[Stream Packet Number: 4]
> [Conversation completeness: Incomplete, DATA (15)]
[TCP Segment Len: 282]
Sequence Number: 1 (relative sequence number)
Sequence Number (raw): 3979841385
[Next Sequence Number: 283 (relative sequence number)]
Acknowledgment Number: 1 (relative ack number)
Acknowledgment number (raw): 3528601433
0101 = Header Length: 20 bytes (5)
> Flags: 0x018 (PSH, ACK)
Window: 515

00
00
00
00
00
00
00
00
00
00
01
01
01
01

4.3 Фильтр dns

dns						
No.	Time	Source	Destination	Protocol	Length	Info
19865	664.205212	172.16.58.120	192.168.150.37	DNS	85	Standard query 0xa548 A 159-223-41-199.1-agent.me
19866	664.222673	192.168.150.37	172.16.58.120	DNS	170	Standard query response 0xa548 No such name A 159-223-41-199.1-agent.me SOA ns-1096.am...
19867	664.227107	172.16.58.120	192.168.150.37	DNS	74	Standard query 0x4254 A perr.l-err.biz
19868	664.228428	192.168.150.37	172.16.58.120	DNS	138	Standard query response 0x4254 A perr.l-err.biz A 45.77.221.213 A 45.63.22.51 A 66.135...
21073	705.771874	172.16.58.120	192.168.150.37	DNS	75	Standard query 0xe101 A perr.l-agent.me
21074	705.774175	192.168.150.37	172.16.58.120	DNS	139	Standard query response 0xe101 A perr.l-agent.me A 45.77.221.213 A 45.63.22.51 A 66.13...
22390	747.828697	172.16.58.120	192.168.150.37	DNS	79	Standard query 0x1aeb A perr.bright-sdk.com
22391	747.830014	192.168.150.37	172.16.58.120	DNS	143	Standard query response 0x1aeb A perr.bright-sdk.com A 66.135.27.97 A 45.77.221.213 A ...
23669	810.563959	172.16.58.120	192.168.150.37	DNS	81	Standard query 0xba7b A 65-20-76-8.1-agent.me
23670	810.600800	172.16.58.120	192.168.150.36	DNS	81	Standard query 0xba7b A 65-20-76-8.1-agent.me
23671	810.602758	192.168.150.37	172.16.58.120	DNS	166	Standard query response 0xba7b No such name A 65-20-76-8.1-agent.me SOA ns-1096.awsdns...
23674	810.632796	192.168.150.36	172.16.58.120	DNS	166	Standard query response 0xba7b No such name A 65-20-76-8.1-agent.me SOA ns-1096.awsdns...
26003	876.283414	172.16.58.120	192.168.150.37	DNS	85	Standard query 0xa817 A 139-59-149-133.1-agent.me
26004	876.303011	192.168.150.37	172.16.58.120	DNS	170	Standard query response 0xa817 No such name A 139-59-149-133.1-agent.me SOA ns-1096.au...
> Frame 16734: Packet, 235 bytes on wire (1880 bits), 235 bytes captured (1880 bi						0000 10 16 54 69 79 63 70 18 a7 60 9c c5 08 00 45 00 tiypc: <-----E
> Ethernet II, Src: Cisco00:9c:c5 (70:18:a7:60:9c:c5), Dst: Intel69:79:63 (f8:1						0010 00 dd 2f ca 00 00 7f 11 cd ef c0 a8 96 25 ac 10 /----- %
> Internet Protocol Version 4, Src: 192.168.150.37, Dst: 172.16.58.120						0020 3a 78 00 35 c8 1d 00 c9 33 c3 04 f5 81 80 00 01 :x-5----- 3-----
> User Datagram Protocol, Src Port: 53, Dst Port: 51229						0030 00 04 00 00 00 00 05 63 74 6c 64 6c 0d 77 69 6e -----c tldl-win
> Domain Name System (response)						0040 64 6f 77 73 75 70 64 61 74 65 03 63 6f 6d 00 00 dousupda te.com--
						0050 01 00 01 c0 0c 00 05 00 01 00 00 00 1c 00 2d 05 -----
						0060 63 74 6c 64 6c 0d 77 69 6e 64 6f 77 73 75 70 64 ctldl-wi ndousupd
						0070 61 74 65 03 63 6f 6d 08 64 65 6c 69 76 65 72 79 ate.com delivery
						0080 09 6d 69 63 72 6f 73 6f 66 74 c0 20 c0 35 00 05 -microso ft- 5--
						0090 00 01 00 00 08 0b 00 1d 08 77 75 2d 62 2d 6e 65 -----wu-b-ne
						00a0 74 0e 74 72 61 66 66 69 63 6d 61 6e 61 67 65 72 t:traffi cmanage
						00b0 03 6e 65 74 00 c0 6e 00 05 00 01 00 00 00 2e 00 -net:n-----
						00c0 1a 02 62 67 09 6d 69 63 72 6f 73 6f 66 74 03 6d -bg mic rosoft:m
						00d0 61 70 06 66 61 73 74 6c 79 c0 06 c0 97 00 01 00 ap-fastl y-----
						00e0 01 00 00 00 1e 00 04 97 65 ce ac -----e--

Рисунок 14: Кадр dns - запрос

4.4 Фильтр dns

dns						
No.	Time	Source	Destination	Protocol	Length	Info
19865	664.205212	172.16.58.120	192.168.150.37	DNS	85	Sta
19866	664.222673	192.168.150.37	172.16.58.120	DNS	170	Sta
19867	664.227107	172.16.58.120	192.168.150.37	DNS	74	Sta
19868	664.228428	192.168.150.37	172.16.58.120	DNS	138	Sta
21073	705.771874	172.16.58.120	192.168.150.37	DNS	75	Sta
21074	705.774175	192.168.150.37	172.16.58.120	DNS	139	Sta
22390	747.828697	172.16.58.120	192.168.150.37	DNS	79	Sta
22391	747.830014	192.168.150.37	172.16.58.120	DNS	143	Sta
23669	810.563959	172.16.58.120	192.168.150.37	DNS	81	Sta
23670	810.600800	172.16.58.120	192.168.150.36	DNS	81	Sta
23671	810.602758	192.168.150.37	172.16.58.120	DNS	166	Sta
23674	810.632796	192.168.150.36	172.16.58.120	DNS	166	Sta
26003	876.283414	172.16.58.120	192.168.150.37	DNS	85	Sta

> Frame 21074: Packet, 139 bytes on wire (1112 bits), 139 bytes captured (1112 bits) on interface 0

> Ethernet II, Src: Cisco_60:9c:c5 (70:18:a7:60:9c:c5), Dst: Intel_69:79:63 (fc:00:14:69:79:63), Length: 139

> Internet Protocol Version 4, Src: 192.168.150.37, Dst: 172.16.58.120

> User Datagram Protocol, Src Port: 53, Dst Port: 51273

Source Port: 53

Destination Port: 51273

Length: 105

Checksum: 0xd040 [unverified]

[Checksum Status: Unverified]

[Stream index: 582]

[Stream Packet Number: 6]

[Timestamps]

UDP payload (97 bytes)

Domain Name System (response)

Рисунок 15: Кадр dns - ответ

4.5 Фильтр quic

quic						
No.	Time	Source	Destination	Protocol	Length	
16124	538.073707	172.16.58.120	64.233.161.105	QUIC	71 B	
16128	538.122018	64.233.161.105	172.16.58.120	QUIC	68 B	
16832	565.076363	172.16.58.120	64.233.161.105	QUIC	71 B	
16833	565.118504	64.233.161.105	172.16.58.120	QUIC	68 B	
17806	592.078465	172.16.58.120	64.233.161.105	QUIC	71 B	
17807	592.123355	64.233.161.105	172.16.58.120	QUIC	68 B	
18245	605.823508	64.233.161.105	172.16.58.120	QUIC	110 B	
32021	1094.340768	172.16.58.120	173.194.222.94	QUIC	1292 B	
32022	1094.340861	172.16.58.120	173.194.222.94	QUIC	1292 B	
32023	1094.342652	172.16.58.120	173.194.222.94	QUIC	121 B	
32024	1094.342910	172.16.58.120	173.194.222.94	QUIC	444 B	
32025	1094.358035	173.194.222.94	172.16.58.120	QUIC	82 B	
32026	1094.358378	173.194.222.94	172.16.58.120	QUIC	1292 B	

> Frame 18245: Packet, 110 bytes on wire (880 bits), 110 bytes captured (880 bits) on interface 0

> Ethernet II, Src: Cisco_60:9c:c5 (70:18:a7:60:9c:c5), Dst: Intel_69:79:63 (fc:00:14:69:79:63)

> Internet Protocol Version 4, Src: 64.233.161.105, Dst: 172.16.58.120

> User Datagram Protocol, Src Port: 443, Dst Port: 56150

Source Port: 443

Destination Port: 56150

Length: 76

Checksum: 0x3cfa [unverified]

[Checksum Status: Unverified]

[Stream index: 433]

[Stream Packet Number: 36]

> [Timestamps]

UDP payload (68 bytes)

> QUIC IETF

5. Анализ handshake протокола TCP в Wireshark

5.1 handshake

quic							
No.	Time	Source	Destination	Protocol	Length	Info	
16124	538.073707	172.16.58.120	64.233.161.105	QUIC	71	Protected Pa	
16128	538.122018	64.233.161.105	172.16.58.120	QUIC	68	Protected Pa	
16832	565.076363	172.16.58.120	64.233.161.105	QUIC	71	Protected Pa	
16833	565.118504	64.233.161.105	172.16.58.120	QUIC	68	Protected Pa	
17806	592.078465	172.16.58.120	64.233.161.105	QUIC	71	Protected Pa	
17807	592.123355	64.233.161.105	172.16.58.120	QUIC	68	Protected Pa	
18245	605.823508	64.233.161.105	172.16.58.120	QUIC	110	Protected Pa	
32021	1094.340768	172.16.58.120	173.194.222.94	QUIC	1292	Initial, DCI	
32022	1094.340861	172.16.58.120	173.194.222.94	QUIC	1292	Initial, DCI	
32023	1094.342652	172.16.58.120	173.194.222.94	QUIC	121	0-RTT, DCID=	
32024	1094.342910	172.16.58.120	173.194.222.94	QUIC	444	0-RTT, DCID=	
32025	1094.358035	173.194.222.94	172.16.58.120	QUIC	82	Initial, SCI	
32026	1094.358378	173.194.222.94	172.16.58.120	QUIC	1292	Initial, SCI	

[Stream index: 433]	0000	f8
[Stream Packet Number: 36]	0010	00
> [Timestamps]	0020	3a
UDP payload (68 bytes)	0030	2f
	0040	4b
✓ QUIC IETF	0050	bc
✓ QUIC Connection information	0060	4a
[Connection Number: 0]		
[Packet Length: 68]		
✓ QUIC Short Header		
0... = Header Form: Short Header (0)		
.1.. = Fixed Bit: True		
..0. = Spin Bit: False		
Remaining Payload: 67ba9bf76c2f0613828be86f75eb708006193168a64bcc384d39222		

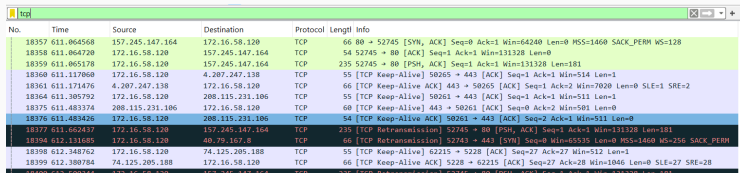
http://info.cern.ch - home of the first website

From here you can:

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- [Browse the first website using the line-mode browser simulator](#)
- [Learn about the birth of the web](#)
- [Learn about CERN, the physics laboratory where the web was born](#)

Рисунок 18: Вторая ступень handshake TCP

5.3 handshake



No.	Time	Source	Destination	Protocol	Length	Info
18357	611.064568	157.245.147.164	172.16.58.120	TCP	66	80 → 52745 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460 SACK_PERM WS=128
18358	611.064720	172.16.58.120	157.245.147.164	TCP	54	52745 → 80 [ACK] Seq=1 Ack=1 Win=131328 Len=0
18359	611.065178	172.16.58.120	157.245.147.164	TCP	235	52745 → 80 [PSH, ACK] Seq=1 Ack=1 Win=131328 Len=181
18360	611.117060	172.16.58.120	4.207.247.138	TCP	55	[TCP Keep-Alive] 50265 → 443 [ACK] Seq=1 Ack=1 Win=514 Len=1
18361	611.171476	4.207.247.138	172.16.58.120	TCP	66	[TCP Keep-Alive ACK] 443 → 50265 [ACK] Seq=1 Ack=2 Win=7020 Len=0 SLE=1 SRE=2
18364	611.305792	172.16.58.120	208.115.231.106	TCP	55	[TCP Keep-Alive] 50261 → 443 [ACK] Seq=1 Ack=1 Win=511 Len=1
18375	611.483374	208.115.231.106	172.16.58.120	TCP	60	[TCP Keep-Alive] 443 → 50261 [ACK] Seq=0 Ack=2 Win=501 Len=0
18376	611.483426	172.16.58.120	208.115.231.106	TCP	54	[TCP Keep-Alive ACK] 50261 → 443 [ACK] Seq=2 Ack=1 Win=511 Len=0
18377	611.662437	172.16.58.120	157.245.147.164	TCP	235	[TCP Retransmission] 52745 → 80 [PSH, ACK] Seq=1 Ack=1 Win=131328 Len=181
18394	612.131685	172.16.58.120	40.79.167.8	TCP	66	[TCP Retransmission] 52743 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
18398	612.348762	172.16.58.120	74.125.205.188	TCP	55	[TCP Keep-Alive] 62215 → 5228 [ACK] Seq=27 Ack=27 Win=512 Len=1
18399	612.380784	74.125.205.188	172.16.58.120	TCP	66	[TCP Keep-Alive ACK] 5228 → 62215 [ACK] Seq=27 Ack=28 Win=1046 Len=0 SLE=27 SRE=28
18400	612.509324	172.16.58.120	157.245.147.164	TCP	235	[TCP Retransmission] 52745 → 80 [PSH, ACK] Seq=1 Ack=1 Win=131328 Len=181

Рисунок 19: Третья ступень handshake TCP

5.4 График потока

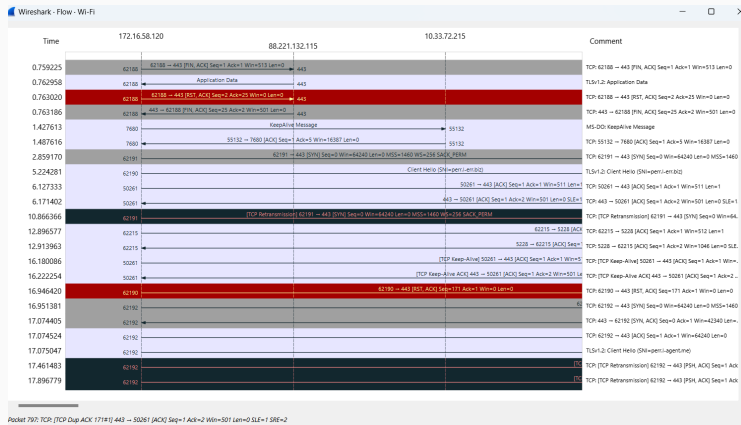


Рисунок 20: График потока

5.5 Вывод

- В результате выполнения работы были изучены посредством Wireshark кадры Ethernet, произведен анализ PDU протоколов транспортного и прикладного уровней стека TCP/IP.